

“WHAT WAS THE USE OF THIS GREAT ENGINE SET AT WORK AGES AGO TO... FURROW AND KNEAD OVER... THE SURFACE OF THE EARTH? THE GLACIER WAS GOD’S GREAT PLOW.”

Louis Agassiz, Swiss geologist, in *Geological Sketches*, 1866



The Marjerie Glacier in Glacier Bay, Alaska, USA, flows out from the mountains to the sea and extends a distance of 21 miles (34 km).

THIS WAS THE YEAR in which telecommunications really began. Three men had already been working on the idea of an electric telegraph—**Samuel Morse** in the US and **William Cooke** (1806–79) and **Charles Wheatstone** (1802–75) in England. In 1837, it became a reality. Cooke and Wheatstone were the first to succeed with their 1.2-mile- (2 km-) long telegraph from Euston to Camden in London. In the end, the simplicity of the Morse telegraph’s single wire system and dot-dash code meant it was adopted as the standard telegraph and the Wheatstone-Cooke model was discarded. An equally momentous breakthrough was the development of the **first**



successful photographic process by French painter **Louis-Jacques-Mandé Daguerre** (1787–1851). Daguerre had long hunted for a means to fix the fleeting images he saw in his artist’s camera obscura. He finally succeeded with his invention of the process known as **daguerreotyping**, which involved capturing a one-off photograph in chemicals coated on a silver-plated copper sheet.

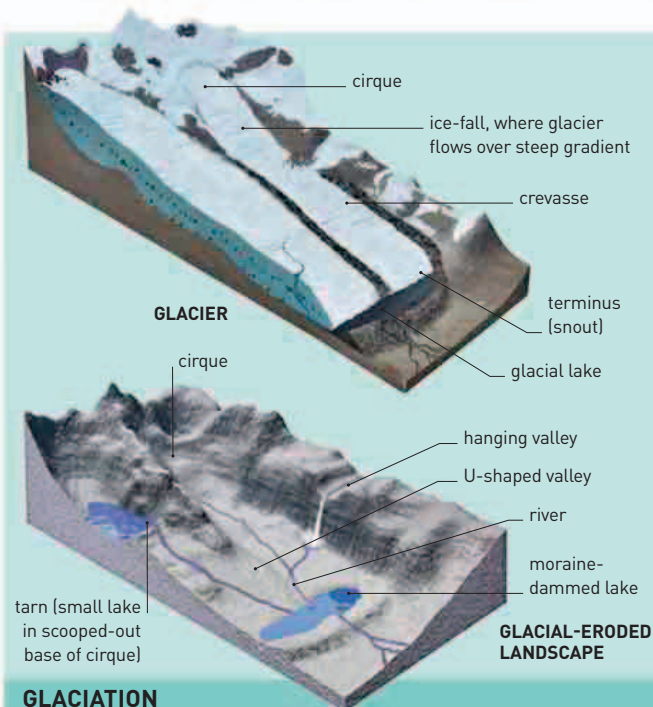
Daguerreotype camera
Dating from the 1840s, this is one of the first cameras made for taking photographs. The plate for each photo is slotted into the back.

His first daguerreotype, taken in 1837, was blurry, but within a few years daguerreotypes were recording photographs of astonishing clarity.

The most extraordinary scientific insight of this year was understood by no one at the time but its creator, British scientist **Michael Faraday**. This was the idea of **fields of force**—regions around magnets and electric currents in which their effects are felt. Faraday argued that in these fields, electric charges are pushed by the movement of invisible **lines of force**, which can be revealed by the pattern iron filings make around a magnet. This is why compasses swivel in magnetic fields and magnets move charged particles to create electric currents.

French mathematician **Siméon Denis Poisson** (1781–1840) developed the valuable statistical idea of **Poisson distribution**: the probability of a given number of events occurring in a particular time if they occur at an average rate.

In Paris, following the discovery of the **green pigment chlorophyll** in plant leaves in



tarn (small lake in scooped-out base of cirque)

GLACIER

GLACIAL-ERODED LANDSCAPE

GLACIATION

Glaciers accumulate such a weight of ice over the years that they possess an immense power to shape the landscape. They can carve out vast U-shaped valleys, cut-off hillsides, and leave behind piles of rock debris. This process of modifying the landscape is known as glacialiation and the landscapes left behind by the glaciers of the ice age are unmistakable to geologists today.

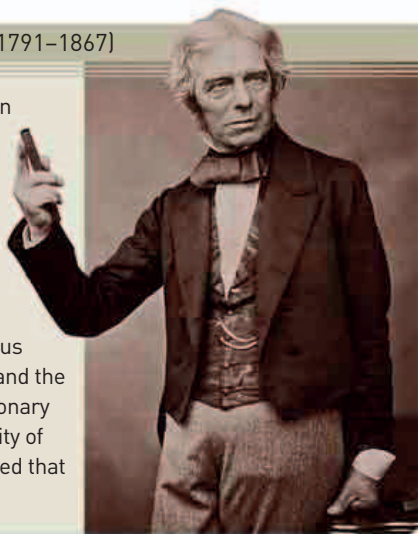
1817, French physiologist **Henri Dutrochet** (1776–1847) argued that chlorophyll is the key to **photosynthesis**, by which plants fix oxygen from the air using sunlight (see 1787–88).

Meanwhile, in Switzerland, geologist **Louis Agassiz**

(1807–73) published *Études sur les Glaciers* (*Studies of Glaciers*), arguing that the **Earth was once subject to an ice age** and that traces of ice erosion and deposition by vast glaciers and ice sheets are still evident in the landscape today.

MICHAEL FARADAY (1791–1867)

The son of a poor London blacksmith, Michael Faraday was taken on as an assistant to chemist Humphry Davy at London’s Royal Institution in 1813. His discoveries in electromagnetism gave us both the electric motor and the generator. He was a visionary theorist who saw the unity of natural forces and showed that light is electromagnetic.



September 6 William Cooke and Charles Wheatstone start the world’s first commercial electric telegraph line in London

Michael Faraday begins to develop the idea of electromagnetic fields

American inventor Samuel Morse constructs his electric telegraph

French artist Louis Daguerre’s first daguerreotype photograph

Swiss geologist Louis Agassiz proposes his theory of glacialiation

French physiologist Henri Dutrochet realizes the role of chlorophyll in photosynthesis

English engineer Isambard Kingdom Brunel’s steamship SS Great Western is launched in Bristol

French mathematician Siméon Denis Poisson introduces Poisson distribution